My AWS Network

VPC: Virtual Private Cloud is a regional resource where you can deploy your resource within the network

Internet Protocal:
- **IPv4**: (total 4.3 billion addresses)
  - Public IPv4: will change within AWS. Not free, only 750 hrs per month free.
  - Private IPv4: does not change
- **IPv6**: (3.4 x 10^38, much more)
  - All public, all free

VPC CIDR Range: A range of IP addresses allowed in the VPC. 10.0.0.0/16

Subnets: One AZ has 2 subnets: 1 public and 1 private. Only public is accessible from the Internet. You can put your databases in private and EC2+load balancer in public. Use **Route Tables** to define access to the internet and between subnets.

Security Groups
- A firewall that controls traffic to and from an **EC2** instance
- Rules include IP addresses and other security groups
- Can have only ALLOW rules
- Stateful: Return traffic is automatically allowed, regardless of any rules

NACL (Network Access Control List)
- A firewall which controls traffic from and to subnet, attached at the **Subnet** Level
- Rules only include IP addresses
- Can have ALLOW and DENY rules
- Stateless: Return traffic must be explicitly allowed by rules

Elastic IP: can attach a fixed public IPv4 address to EC2

Internet Gateway: connects VPC to the internet and thus makes that subnet public.

NAT Gateway or NAT Instances: allow instances in Private subnets to access the internet while remaining private

VPC Peering
- Connect 2 VPC, privately using AWS network, make them as if in the same network
- 2 VPCs' CIDRs (IP addresses) must not overlap
- Connection is **not transitive**

Transit Gateway
- For having transitive peering between thousands of VPC and on-premises, hub-and-spoke (star) connection
- One single Gateway to provide this functionality
- Works with Direct Connect Gateway,VPN connections

VPC Endpoints:
- Allow to connect to AWS Services using a private network instead of public www network for enhanced security and low latency
- **VPC Endpoint Gateway** is for only **S3** and **DynamoDB**
- **VPC Endpoint Interface** is for most services like Cloudwatch including S3 and DynamoDB

VPC Flow Logs
- Capture info about IP traffic
  - VPC Flow Logs
  - Subnet Flow Logs
  - Elastic Network Interface Flow Logs
- Helps to monitor & troubleshoot connectivity issues within VPC and to Internet
- Capture network info from AWS managed services
- go to S3, CloudWatch Logs, Data Firehose

2. Direct Connect (DX)
- Establish a physical connection between on-premises and AWS
- Goes over private network
- Takes at least a month to establish
- The connection is private, secure, and fast

1. Site-to-Site VPN
- Connect on-premise VPN to AWS
- The connection is automatically encypted
- Goes over public Internet
- On-Premises: Use **Customer Gateway (CGW)**
- AWS: Use **Virtual Private Gateway (VGW)**
- Faster setup, less secure

AWS Client VPN
- Connect your computer using **OpenVPN tp** your private network in AWS and on-premises
- Goes over the public Internet
- Can connect your EC2 instances over a private IP as if you were in the private VPC network

AWS PrivateLink (VPC Endpoint Services)
- Most secure and scalable way to expose a service to 1000s of VPCs
- No peering, internet gateway, NAT, route tables
- Need a **Network load balancer** (Service VPC) and **ENI** (Client VPC)